Chapter 16: Lazy Loading

1. Overview

Lazy loading is the practice of deferring the download and execution of JavaScript until it is actually needed, instead of shipping one enormous bundle that the browser must parse before the app becomes interactive. In Angular, lazy loading is primarily a **router feature**: instead of eagerly importing every feature's module or component at bootstrap, the router is configured with a route whose component/module is fetched on demand via a dynamic `import()` call, the moment the user navigates to a matching URL.

Historically (Angular ≤15, NgModule-based apps) this meant `loadChildren` pointing at a lazy-loaded `NgModule`. Since Angular 14+ introduced standalone components, and especially from Angular 15/17 onward as standalone became the default, the idiomatic mechanism is `loadComponent` (for a single routed component) or `loadChildren` returning an array of standalone `Routes` (for a lazy feature area with its own child routes). Angular 15 also added standalone `loadChildren` for child route configs without needing an NgModule wrapper at all.

Why it matters for interviews: lazy loading sits at the intersection of the **Router**, the **build system (esbuild/Webpack via Angular CLI)**, and **runtime dependency injection (environment injectors)**. A candidate who can explain *how* a route becomes a separate chunk, *how* the router's environment injector isolates lazy providers, and *why* preloading strategies exist demonstrates real production experience, not just tutorial familiarity.

Core goals of this chapter:

- Contrast `loadChildren` (module-based, legacy) with `loadComponent` and standalone `loadChildren` (modern).
- Explain preloading strategies (`NoPreloading`, `PreloadAllModules`, custom) and when each is appropriate.
- Show how route-level code splitting produces separate chunks and how that maps to `ng build` output.
- Cover lazy-loading non-routed code via raw `import()` (e.g., heavy libraries, PDF/chart generators).
- Explain bundle analysis tooling (`source-map-explorer`, `webpack-bundle-analyzer`, Angular's built-in stats) to verify lazy loading is actually working.

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2. Core Concepts

2.1 `loadChildren` — the legacy NgModule pattern

Before standalone components, every lazy route pointed at an `NgModule`:

```
// app-routing.module.ts
const routes: Routes = [
  {
    path: 'admin',
    loadChildren: () => import('./admin/admin.module').then(m => m.AdminModule)
  }
];
```

- `loadChildren` accepts a function returning a `Promise`. The function is not called until the router actually needs to activate that route (or a preloading strategy decides to fetch it early).
- The imported `AdminModule` declares its own routes (typically via `RouterModule.forChild(adminRoutes)`), its own components, and can register its own providers.
- The Angular CLI's builder statically analyzes `loadChildren` strings/arrow-function imports and automatically splits `AdminModule` and everything it exclusively depends on into a separate output chunk (`admin-module.js` or a content-hashed equivalent).
- **Downside**: NgModules add ceremony — you need a module class, a routing module, `declarations`, `exports`, etc., purely to get code-splitting. This is why standalone lazy loading replaced it as the default recommendation from Angular 15 onward (and Angular 19's schematics no longer scaffold NgModules by default).

2.2 `loadComponent` — standalone component lazy loading

For a route that renders a single standalone component (no nested child routes needed), `loadComponent` is the direct, minimal-ceremony replacement:

```
const routes: Routes = [
  {
    path: 'profile',
    loadComponent: () => import('./profile/profile.component').then(c => c.ProfileComponent)
  }
];
```

- No module wrapper required. The component must be `standalone: true` (implicit default from Angular 19) with its own `imports` array for template dependencies.
- The CLI still splits this into its own chunk because the dynamic `import()` is statically analyzable.
- `loadComponent` can be combined with `title`, `canActivate`, `resolve`, etc., exactly like any other route.

2.3 Standalone `loadChildren` — lazy feature areas without NgModule

When a feature area needs its **own child routes** (not just one component), you no longer need an NgModule — `loadChildren` can resolve directly to a `Routes` array:

```
// admin.routes.ts
export const ADMIN_ROUTES: Routes = [
  { path: '', component: AdminDashboardComponent },
```

```
  { path: 'users', component: UsersComponent },
  { path: 'users/:id', component: UserDetailComponent },
];

// app.routes.ts
const routes: Routes = [
  {
    path: 'admin',
    loadChildren: () => import('./admin/admin.routes').then(m => m.ADMIN_ROUTES)
  }
];
```

This is the modern equivalent of feature-module lazy loading: same chunking behavior, same route-nesting capability, zero NgModule boilerplate. Providers scoped to the feature are supplied via the route's own `providers` array (see 2.5) instead of an `NgModule.providers` array.

2.4 loadComponent vs loadChildren — decision table

Concern	`loadComponent`	`loadChildren` **(standalone routes or NgModule)**
Use case	Single routed component, no children	Feature area with multiple sub-routes
Child routing	Not supported directly on that route	Fully supported (`children` array in the imported routes)
Providers scoping	Via route's own `providers`	Via feature routes' `providers`, or (legacy) `NgModule.providers`
Chunk granularity	One chunk per component	One chunk per feature area (can bundle many components)
Boilerplate	Minimal	Minimal (standalone) / heavy (NgModule)

A common interview trap: **you can combine both** — a `loadChildren` entry resolving to standalone routes, where some of those child routes themselves use `loadComponent` for further splitting. This produces a two-level lazy tree: the feature shell chunk loads first, and individual heavy components inside it split further.

2.5 Route-level providers (environment injector scoping)

Since Angular 14.1+, routes accept a `providers` array that creates a new **environment injector** scoped to that route (and its children), regardless of whether the route uses `loadComponent` or `loadChildren`:

```
{
  path: 'admin',
  loadChildren: () => import('./admin/admin.routes').then(m => m.ADMIN_ROUTES),
  providers: [
    { provide: ADMIN_CONFIG, useValue: { pageSize: 50 } },
    AdminApiService
  ]
}
```

This is the direct standalone replacement for "provide a service in a lazy NgModule so it gets its own instance" — a very common legacy interview topic (see Edge Cases).

2.6 Preloading Strategies

Lazy loading alone means the chunk downloads only on navigation — the user pays a network round-trip delay the first time they click a lazy link. **Preloading** lets Angular fetch lazy chunks *in the background after the initial app bootstrap*, so by the time the user navigates, the chunk is already cached, giving instant navigation without bloating the initial bundle.

Configured via the second argument to `provideRouter` (standalone) or `RouterModule.forRoot`:

```
// standalone bootstrap
import { provideRouter, withPreloading, PreloadAllModules } from '@angular/router';

bootstrapApplication(AppComponent, {
  providers: [
    provideRouter(routes, withPreloading(PreloadAllModules))
  ]
});
```

```
// NgModule style
RouterModule.forRoot(routes, { preloadingStrategy: PreloadAllModules })
```

Built-in strategies:

- `NoPreloading` (default): lazy chunks load strictly on-demand, at the moment of navigation. Smallest initial network usage, slowest first navigation to each lazy route.
- `PreloadAllModules`: immediately after the initial navigation completes, the router walks the full route config and preloads *every* lazy route's chunk, sequentially, using the browser idle time. Great for small-to-medium apps where you want snappy in-app navigation and don't mind extra background bandwidth. Bad for apps with rarely-used, huge lazy sections (e.g., an admin panel 99% of users never open) or for users on metered/slow connections.
- **Custom strategy**: implement `PreloadingStrategy` interface to selectively preload based on route `data`, network conditions (`navigator.connection.effectiveType`), user role, etc.

2.7 Custom Preloading Strategy — mechanics

A `PreloadingStrategy` is a class implementing one method:

```
interface PreloadingStrategy {
  preload(route: Route, load: () => Observable<any>): Observable<any>;
}
```

- `route` — the `Route` config object being considered (includes `.data`, `.path`, `.loadChildren`, etc.).
- `load` — a function that, when called, triggers the actual dynamic import and returns an `Observable` that emits when loading finishes.
- Returning `load()` triggers preloading; returning `of(null)` (or any observable that doesn't call `load`) skips preloading for that route.

This is invoked by the router **once per lazy route**, right after the initial navigation resolves — see Internal Working for the exact sequencing.

2.8 Route-level Code Splitting — what "route-level" means

"Code splitting" is a bundler-level concept: instead of one `main.js`, the build output contains multiple JS files ("chunks"), each loaded independently. Angular's router integrates with this by ensuring every distinct `import()` expression used in `loadChildren` / `loadComponent` becomes a **split point** the bundler (esbuild in the modern `@angular/build:application` builder, or Webpack in the legacy builder) recognizes and extracts into its own chunk, with the exact chunk boundary determined by module dependency graph analysis (shared dependencies get hoisted into common chunks to avoid duplication).

Key nuance: **route-level** granularity means one chunk per lazy route entry point, not one chunk per component within that route unless you further split with nested `loadComponent`. If `AdminModule` / `ADMIN_ROUTES` eagerly imports ten components at the top of the file, all ten ship in the single `admin` chunk — splitting stops at the `import()` boundary you actually wrote.

2.9 Lazy-loading Non-Routed Code — dynamic `import()`

Not everything that should be deferred is a route. Common candidates: charting libraries (Chart.js, d3), PDF generators, rich text editors, heavy validation/crypto libraries, polyfills for a legacy browser branch — code needed only when a specific button is clicked or a specific condition is true, unrelated to routing.

```
async exportToPdf() {
  const { jsPDF } = await import('jspdf');  // only downloaded when this method actually
runs
  const doc = new jsPDF();
  doc.text('Report', 10, 10);
  doc.save('report.pdf');
}
```

- The Angular CLI/esbuild recognizes *any* dynamic `import()` in application code (not just router config) as a split point, producing a separate lazy chunk.
- Works identically inside standalone components, services, directives — anywhere in TS code.
- Common pattern: wrap in a signal/flag so the import only fires once (memoize the promise) if the action can be triggered repeatedly.
- Angular also has `@defer` blocks (Angular 17+) in templates, which are a *declarative*, template-driven wrapper around this same dynamic-import mechanism for view-level lazy loading (deferred rendering

based on viewport, interaction, idle, timer, or custom conditions) — worth mentioning as the modern alternative to hand-rolled `import()` for template-bound lazy UI, though it's commonly treated as its own chapter topic.

2.10 Bundle Analysis

You cannot verify lazy loading is effective just by reading code — you must inspect the actual build output.

Tools:

- `ng build --stats-json` — emits `stats.json` alongside the build, describing every chunk, its modules, and sizes.
- `source-map-explorer dist/<project>/browser/*.js` — visual treemap of what's inside each output chunk, using source maps to attribute bytes back to original source files/node_modules packages. Extremely effective at catching "why is my lazy chunk 800KB" surprises (e.g., accidentally importing a barrel file that pulls in the whole app).
- `webpack-bundle-analyzer dist/<project>/stats.json` — interactive treemap UI (works with the legacy Webpack-based builder or with `--stats-json` output).
- Angular CLI's own build summary (printed after every `ng build --configuration production`) — lists initial chunk sizes and **lazy chunk sizes** with budgets; CI can be configured (`angular.json` → `budgets`) to fail the build if any chunk exceeds a size threshold, which is the most common way teams catch lazy-loading regressions automatically.

What to look for:

- The **initial bundle** (`main.js`, `polyfills.js`, `styles.css`, and any eagerly-loaded vendor chunk) should be as small as possible — this is what blocks First Contentful Paint / Time to Interactive.
- Each **lazy chunk** should roughly correspond 1:1 with a `loadChildren` / `loadComponent` call; if two unrelated lazy routes appear merged into one chunk, it usually means they share a common eagerly-imported module that itself imports both — check for accidental cross-imports between feature areas.
- Watch for a library imported both eagerly (in `main.js`) and lazily (duplicated in a lazy chunk) — that indicates it wasn't properly hoisted to a shared chunk, often due to differing import specifiers or a barrel export re-exporting it from two places.

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3. Code Examples

3.1 `loadComponent` for a standalone route

```
// app.routes.ts
import { Routes } from '@angular/router';

export const routes: Routes = [
  { path: '', pathMatch: 'full', redirectTo: 'home' },
  {
    path: 'home',
    loadComponent: () => import('./home/home.component').then(m => m.HomeComponent)
  },
  {
```

```
    path: 'profile',
    loadComponent: () =>
      import('./profile/profile.component').then(m => m.ProfileComponent),
    // route-scoped providers - fresh environment injector just for this route subtree
    providers: [ProfileFacadeService]
  }
];
```

```
// profile.component.ts
import { Component } from '@angular/core';
import { CommonModule } from '@angular/common';

@Component({
  selector: 'app-profile',
  standalone: true,
  imports: [CommonModule],
  template: `<h2>Profile for {{ name }}</h2>`
})
export class ProfileComponent {
  name = 'Ada Lovelace';
}
```

3.2 Standalone `loadChildren` for a full lazy feature area

```
// admin/admin.routes.ts
import { Routes } from '@angular/router';
import { AdminApiService } from './admin-api.service';

export const ADMIN_ROUTES: Routes = [
  {
    path: '',
    providers: [AdminApiService], // scoped to whole admin subtree
    children: [
      {
        path: '',
        loadComponent: () =>
          import('./dashboard/dashboard.component').then(m => m.DashboardComponent)
      },
      {
        path: 'users',
        loadComponent: () =>
          import('./users/users.component').then(m => m.UsersComponent)
      },
      {
        path: 'users/:id',
        loadComponent: () =>
          import('./users/user-detail.component').then(m => m.UserDetailComponent)
      }
    ]
  }
];
```

```
// app.routes.ts
export const routes: Routes = [
  {
    path: 'admin',
    canActivate: [authGuard],
    loadChildren: () => import('./admin/admin.routes').then(m => m.ADMIN_ROUTES)
  }
];
```

This produces one top-level `admin` chunk (containing the shared `AdminApiService` and route wiring), plus further split-off chunks for `dashboard`, `users`, and `user-detail` since each uses its own `loadComponent`.

3.3 Legacy NgModule `loadChildren` (for contrast / maintaining older codebases)

```
// admin.module.ts
@NgModule({
  declarations: [DashboardComponent, UsersComponent],
  imports: [
    CommonModule,
    RouterModule.forChild([
      { path: '', component: DashboardComponent },
      { path: 'users', component: UsersComponent }
    ])
  ],
  providers: [AdminApiService] // instantiated once, in this module's injector
})
export class AdminModule {}
```

```
// app-routing.module.ts
const routes: Routes = [
  { path: 'admin', loadChildren: () => import('./admin/admin.module').then(m =>
m.AdminModule) }
];
```

3.4 Custom Preloading Strategy

A strategy that preloads only routes explicitly opted in via route `data`, and additionally skips preloading entirely on slow/metered connections:

```
import { Injectable } from '@angular/core';
import { PreloadingStrategy, Route } from '@angular/router';
import { Observable, of } from 'rxjs';

@Injectable({ providedIn: 'root' })
export class SelectivePreloadingStrategy implements PreloadingStrategy {

  preload(route: Route, load: () => Observable<any>): Observable<any> {
    // 1. Respect explicit opt-in flag on the route config
    if (!route.data?.['preload']) {
```

```
      return of(null);
    }

    // 2. Respect the user's connection quality (Network Information API)
    const connection = (navigator as any).connection;
    if (connection?.saveData === true) {
      return of(null); // user explicitly asked to save data
    }
    if (connection?.effectiveType && /2g/.test(connection.effectiveType)) {
      return of(null); // too slow to bother preloading in background
    }

    // 3. Otherwise trigger the actual dynamic import
    console.log(`Preloading: ${route.path}`);
    return load();
  }
}
```

```
// route config opts in per-route
{
  path: 'reports',
  loadChildren: () => import('./reports/reports.routes').then(m => m.REPORTS_ROUTES),
  data: { preload: true }
}
```

```
// bootstrap wiring (standalone)
bootstrapApplication(AppComponent, {
  providers: [
    provideRouter(routes, withPreloading(SelectivePreloadingStrategy))
  ]
});
```

```
// NgModule wiring (legacy)
@NgModule({
  imports: [RouterModule.forRoot(routes, { preloadingStrategy: SelectivePreloadingStrategy
})]
})
export class AppRoutingModule {}
```

3.5 Lazy-loading a non-routed heavy library

```
import { Component, signal } from '@angular/core';

@Component({
  selector: 'app-chart-panel',
  standalone: true,
  template: `
    <button (click)="renderChart()">Show Chart</button>
    <canvas #canvas></canvas>
  `
```

```
})
export class ChartPanelComponent {
  private chartLoaded = signal(false);

  async renderChart() {
    if (this.chartLoaded()) return;
    // 'chart.js' is not in main.js; it ships as its own chunk, fetched here on demand
    const { Chart, registerables } = await import('chart.js');
    Chart.register(...registerables);
    // ... instantiate chart using canvas ref
    this.chartLoaded.set(true);
  }
}
```

3.6 Deferred/memoized dynamic import (avoid re-fetching on repeat calls)

```
export class PdfExportService {
  private jspdfModule?: Promise<typeof import('jspdf')>;

  private loadJsPdf() {
    // memoize so the import() promise, and the network fetch, only happens once
    this.jspdfModule ??= import('jspdf');
    return this.jspdfModule;
  }

  async export(content: string) {
    const { jsPDF } = await this.loadJsPdf();
    const doc = new jsPDF();
    doc.text(content, 10, 10);
    doc.save('export.pdf');
  }
}
```

4. Internal Working

4.1 How the build system creates separate chunks

1. **Static analysis of `import()`.** The Angular CLI's application builder (esbuild-based since Angular 17's default `@angular/build:application`, formerly Webpack) treats every syntactic dynamic `import('...')` expression — whether inside `loadChildren`, `loadComponent`, or plain application code — as a **split point**. This is a standard ECMAScript dynamic import, so ordinary bundler code-splitting rules apply; Angular's compiler doesn't need special-case logic for `loadChildren`/`loadComponent` beyond ensuring the router *calls* that function lazily at runtime.
2. **Module graph partitioning.** The bundler builds a full dependency graph starting from `main.ts`. Every module reachable only through a dynamic import (and not also reachable eagerly) is placed in its own chunk (or a shared chunk if multiple lazy entry points both depend on it, to avoid duplicate bytes — the bundler hoists common dependencies into a shared chunk fetched once and cached).

3. **Emit as named/hashed files.** Output chunks get content-hashed filenames (e.g., `chunk-XYZAB123.js`) for cache-busting; the entry chunk (`main.js`) contains a runtime chunk-loading manifest mapping internal module IDs to chunk URLs.
4. **Angular-specific compilation still runs per chunk.** Each component's template is still AOT-compiled into its render function ahead of time — code-splitting only affects *when the JS is fetched/executed*, not whether Ivy compilation happened at build time. Lazy chunks contain already-compiled component factories, not raw templates needing runtime compilation.

4.2 How the router resolves and instantiates a lazy route at runtime

1. **Navigation starts.** `Router.navigateByUrl()`/link click triggers the navigation pipeline: URL parsing → recognized against the `Routes` config → guards (`canActivate`, `canMatch`) run.
2. **Route match found with `loadChildren`/`loadComponent`.** The router does **not** eagerly call every route's loader at config time — the function reference just sits there until matched. When the URL segment matches, the router invokes the stored loader function, which executes the dynamic `import()`, returning a `Promise`.
3. **Browser fetches the chunk.** This is a real network request (or a cache hit if previously preloaded/visited). The navigation pipeline suspends at this point (observables from the loader are awaited as part of the router's RxJS pipeline) — this is why users can see a brief blank/loading state on first navigation to a lazy route unless a route-level resolver/loading indicator is used.
4. **Module/component resolved from the promise.** For `loadComponent`, the resolved value is the component class itself, already `@Component`-decorated. For `loadChildren` resolving to `Routes`, the returned array is spliced into the router's in-memory route config as children of the matched path (this is why children of a lazy route only "exist" from the router's perspective after the chunk has loaded once).
5. **A new environment injector is created.** This is the crux of the DI story: if the matched route (or its lazy children) declares `providers: [...]`, or — legacy case — resolves to an `NgModule` with its own `providers`, Angular creates a **new** `EnvironmentInjector` as a child of the root environment injector, registers those providers into it, and associates it with that router state snapshot's `ActivatedRouteSnapshot`. Any component instantiated under that route resolves its constructor dependencies through this child injector first, falling back up the injector tree (child → parent → root → platform) if not found locally. This is exactly how `providedIn: 'root'` singletons remain a single instance app-wide, while route-scoped providers give you a fresh instance per lazy-loaded subtree.
6. **Component instantiation and view creation.** With the class resolved and its injector ready, Angular instantiates the component (or, for module-based lazy children, first bootstraps the module's injector context then the matched component within it), runs change detection, and completes the navigation (`NavigationEnd`).
7. **Subsequent navigations to the same route reuse the cached chunk** (browser HTTP cache / module registry) — the dynamic `import()` promise for an already-loaded specifier resolves instantly without a new network round trip, and the router does not recreate the environment injector unless the route is fully destroyed and re-matched (e.g., navigating away to an unrelated route and back, depending on route reuse strategy).

4.3 Where preloading fits into this sequence

`PreloadingStrategy` is invoked by the router's `RouterPreloader` service, which subscribes to the *first* successful `NavigationEnd` event after bootstrap. At that point it walks the entire route config recursively and, for every route with a `loadChildren` / `loadComponent`, calls the configured strategy's `preload(route, load)`. If the strategy returns `load()`, the same dynamic-import machinery from 4.1–4.2 runs **immediately**, in the background, *without* creating the environment injector or instantiating any component yet — only the chunk is fetched and cached; the actual injector creation and component instantiation from step 5–6 above still only happens when the user actually navigates there.

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5. Edge Cases & Gotchas

5.1 Duplicate service instances from lazy-scoped providers

The single most common lazy-loading interview trap: a service provided in a lazy module/route's `providers` array gets **a separate instance** from the one you might expect to be a singleton.

```
// shared.service.ts
@Injectable() // NOTE: no providedIn: 'root'
export class SharedService { count = 0; }

// admin.module.ts / admin.routes.ts
providers: [SharedService]   // <- creates a NEW instance scoped to this lazy subtree
```

If `SharedService` is *also* listed in the root injector's providers (or another eager module), you now have two independent instances — state set in one is invisible in the other. This bites teams migrating from eager to lazy loading: a service that worked fine as an app-wide singleton silently forks the moment its owning module becomes lazy, because NgModule/route-level `providers` always create a child injector, they never "merge" into root.

Fix / best practice: default to `@Injectable({ providedIn: 'root' })` for anything meant to be a true singleton — this uses Angular's tree-shakable DI where the service is registered directly against the root injector regardless of which module first imports it, immune to this scoping trap. Only use route/module-level `providers` when you deliberately *want* a fresh instance per lazy subtree (e.g., a wizard-flow state service that should reset each time the wizard route subtree is entered).

5.2 Preloading strategy timing

- `PreloadAllModules` starts preloading only **after the first navigation completes**, not at raw app bootstrap — so it never competes with the critical initial render, but it does mean a user who navigates extremely fast (or has an extremely slow first navigation) might still hit a non-preloaded chunk.
- Preloading is **sequential/sequenced through RxJS merge**, not necessarily one-at-a-time; `PreloadAllModules` uses `mergeMap` internally so multiple chunks can be in flight concurrently — on constrained connections this can cause preloading to compete with in-app data requests (XHR/fetch for the current page's actual data) for bandwidth, ironically slowing down the *current* page.
- Preloading only fetches and caches the **JS chunk** — it does not run route guards, does not create the environment injector, and does not instantiate components. If a route has an expensive `canActivate`

guard performing a network call, preloading does not trigger it; that cost is still paid at actual navigation time. Candidates sometimes mistakenly claim preloading "pre-warms" guards/resolvers — it does not.

- Custom strategies that key off `route.data` must remember `data` is only visible on the exact `Route` object as authored — nested child routes need `data` set at each level if you want that level individually toggled.

5.3 Circular lazy imports

If `feature-a.routes.ts` lazily imports something that (directly or via a shared barrel file) imports back into `feature-b`, and `feature-b` lazily imports back toward `feature-a`, you can end up with:

- Both chunks bundling large overlapping code because the bundler can't cleanly separate a true cycle into two independent lazy chunks — it may fall back to merging them into one larger chunk, defeating the purpose of the split, or duplicate the shared middle ground into both chunks (bundle bloat) depending on bundler heuristics.
- In pathological cases, a **runtime deadlock-like symptom** isn't literal here (ESM dynamic imports don't deadlock the way synchronous CommonJS circular requires can), but you can get partially-initialized modules if a barrel `index.ts` re-exports from both feature areas and one references a class from the other before its static class fields have finished initializing — manifesting as "cannot read property of undefined" only in production builds where the module evaluation order differs from dev server order.
- **Mitigation:** never let two lazy feature areas import from each other directly. Shared code (models, truly shared UI, shared services) must live in a common module that both eagerly depend on (so it lands in a shared/common chunk or the main bundle), never inside one feature importing from a sibling feature's folder. Lint rules (e.g., a custom ESLint boundary rule or Nx module boundaries) are commonly used to enforce this at the folder-structure level so the mistake is caught in CI, not discovered as a bundle-analyzer anomaly.

5.4 Other frequently-tested gotchas

- `loadComponent` **cannot have** `children`**.** If you need nested routing under a single lazily-loaded component, you must use `loadChildren` with a routes array (even if that array's root path is `''` pointing to your "shell" component) — a route with `loadComponent` and a sibling `children` array is invalid because there's no natural mount point.
- **Guards can be lazy too.** `canActivate` / `canMatch` / `resolve` accept the same `() => import(...)` pattern for functional guards, letting you keep large guard logic (e.g., a heavy permission-tree evaluator) out of the main bundle as well; this is separate from, and composes with, `loadChildren` / `loadComponent`.
- `canMatch` **vs** `canActivate` **for lazy routes.** `canMatch` runs *before* the router commits to a route match at all (useful to skip loading a chunk entirely — e.g., feature-flagged or role-gated features — and fall through to a different route with the same path), whereas `canActivate` runs after the match, so the chunk is typically already being fetched/has fetched by the time `canActivate` executes. Using `canMatch` to gate a feature avoids ever downloading its chunk for users without access.
- **Absolute vs relative dynamic import paths.** `import('./feature/feature.routes')` must be a **statically analyzable string literal** (possibly with limited template-literal variable interpolation for multiple similar chunks) — fully dynamic runtime-computed paths (`import(someVariable)`) defeat static analysis and the bundler cannot split them into a separate chunk (or, in the worst case, bundles the entire possible directory).

- **SSR considerations.** With Angular Universal/SSR, lazy `loadComponent` / `loadChildren` still works, but the server must actually execute the dynamic import during server-side rendering of that route (Node `import()`), and hydration on the client re-triggers the same lazy fetch unless the chunk is already cached from the initial SSR-served HTML's inlined critical scripts — a frequent source of "works fine client-side-only, breaks under SSR" bugs when a lazily-imported library depends on browser-only globals (`window`, `document`) that don't exist during server execution.

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6. Interview Questions & Answers

Q1. What is lazy loading in Angular, and what problem does it solve?
Lazy loading defers downloading and executing a portion of the application's JavaScript until it's actually needed (typically, until the user navigates to a route that requires it), instead of bundling the entire app into one eagerly-loaded file. It solves the problem of large initial bundle size, which directly hurts Time to Interactive and First Contentful Paint — users pay only for the code the routes they actually visit require.

Q2. What's the difference between `loadChildren` and `loadComponent`?
`loadComponent` lazily loads a single standalone component for a route with no further nested routing at that node. `loadChildren` lazily loads either an `NgModule` (legacy) or a standalone `Routes` array (modern), and is used when the lazy feature area needs its own set of child routes. Both use the same underlying mechanism — a function returning a dynamic `import()` promise — the difference is purely in what shape of thing is resolved (a component class vs. a module/routes array) and therefore whether nested child routing is possible.
Interviewer intent: checking whether the candidate conflates "lazy loading" with "NgModules" — many candidates only know the pre-standalone story and can't explain the modern equivalent.

Q3. How do you configure preloading so that all lazy routes are fetched shortly after the app loads?
Pass `withPreloading(PreloadAllModules)` to `provideRouter` (standalone bootstrap) or `{ preloadingStrategy: PreloadAllModules }` to `RouterModule.forRoot` (NgModule bootstrap). This makes the router's `RouterPreloader` walk every lazy route in the config and begin background-fetching each one immediately after the first navigation completes, so subsequent in-app navigations feel instant while the initial bundle stays small.

Q4. When would `PreloadAllModules` be a bad choice?
When the app has large, rarely-visited lazy sections (e.g., an admin console 95% of users never open), or when a significant fraction of users are on slow/metered connections. Preloading everything wastes bandwidth downloading code that may never be used, and can compete with the current page's real data requests for network bandwidth. In these cases, a custom `PreloadingStrategy` that selectively preloads based on route `data` flags, user role, or connection quality (via the Network Information API) is preferable, or simply sticking with `NoPreloading` (the default).

Q5. Write (verbally describe) a custom preloading strategy that only preloads routes explicitly flagged for it.
Implement `PreloadingStrategy`'s `preload(route, load)` method: check `route.data?.['preload']`; if falsy, return `of(null)` (skip); if truthy, return `load()` (trigger the dynamic import). Register routes with `data: { preload: true }` on the ones you want proactively fetched, and wire the class in via `withPreloading(MyStrategy)`. *(See section 3.4 for full code.)*

Q6. Does a `PreloadingStrategy` run the route's guards or instantiate its components?
No. Preloading only triggers the dynamic `import()` to fetch and cache the JS chunk. It does not evaluate `canActivate`/`canMatch` guards, does not run resolvers, does not create the route's environment injector, and does not instantiate any component. All of that still happens only when the user actually navigates to the route. This is a frequent misconception — preloading is purely a network/bundle optimization, not a "pre-warm everything" mechanism.
Interviewer intent: distinguishes candidates who've only read the strategy's name from those who understand the router pipeline's actual phases.

Q7. A service is provided in a lazy-loaded feature's route `providers` array. Another eager part of the app injects the "same" service and expects shared state. What goes wrong, and why?
Because the service isn't registered with `providedIn: 'root'`, providing it again in the lazy route's `providers` array creates a brand-new instance in a child `EnvironmentInjector` scoped to that route subtree — Angular's DI resolves dependency requests by walking up the injector tree starting from the requesting component's own injector, so anything instantiated under the lazy route gets the lazy-scoped instance, while anything outside gets whatever instance the root/eager injector provides (or none, if it was never provided there, causing a `NullInjectorError` instead). The two are entirely independent objects; mutating state in one is invisible to the other. Fix: use `@Injectable({ providedIn: 'root' })` for true app-wide singletons, and reserve route/module-level `providers` for state that should genuinely be fresh per lazy subtree.

Q8. How does the Angular CLI decide what code goes into which chunk?
It relies on the underlying bundler's (esbuild's, in the modern `@angular/build:application` builder) static analysis of the module dependency graph starting from `main.ts`. Every dynamic `import()` expression — whether from router config or plain code — is a split point; any module reachable *only* through such an import is placed in its own chunk, while a module reachable both eagerly and lazily (or from multiple lazy entry points) is hoisted into a shared chunk to avoid duplicating its bytes across multiple output files. Angular doesn't need router-specific bundler logic — the router's job is just to defer *calling* the loader function until runtime; the compiler-level chunking is generic dynamic-import-based code splitting.

Q9. What is a route-scoped environment injector, and when is it created for a lazy route?
It's a child `EnvironmentInjector`, created by the router when a matched route (or the routes/module it lazily resolves to) declares its own `providers`. It's created at the moment the route is actually activated (after its chunk has finished loading), not at chunk-fetch time. Providers registered there are visible to every component instantiated under that route, and DI lookups from those components check this injector before falling back to its parent (eventually reaching the root/platform injector). This gives lazy features a natural way to scope services to "only exists while this feature is active" without polluting the app-wide root injector.

Q10. Explain the full lifecycle of a user clicking a link to a route configured with `loadChildren`.
(1) Router parses the URL and matches it against the `Routes` config; any `canMatch` guards run first and can reject the match before anything loads. (2) Passing `canActivate` guards run next. (3) The router invokes the stored `loadChildren` function, which executes a dynamic `import()` — a real network fetch unless the chunk is already cached (e.g., from prior preloading or a previous visit). (4) The resolved value (an `NgModule` or a `Routes` array) is merged into the router's in-memory config as children of the matched node. (5) If that config declares its own `providers`, a new child `EnvironmentInjector` is created. (6) The matched component is instantiated using that injector for its dependencies, its view is created, and the navigation completes (`NavigationEnd` fires). On a subsequent visit to the same route, step 3's import resolves instantly from the module cache, and step 5's injector may be reused or recreated depending on the route reuse strategy and whether the route was fully destroyed.
Interviewer intent: tests whether the candidate can sequence guard execution, network fetch, DI setup, and

component instantiation correctly — a common area where candidates hand-wave "the router loads it" without the underlying steps.

Q11. Two unrelated lazy feature areas end up bundled into the same output chunk. What are the likely causes, and how would you diagnose it?
Likely causes: (a) both features import a shared module/barrel file, and that shared file itself imports (even transitively) something from the other feature, creating an effective cross-dependency the bundler can't cleanly separate; (b) one feature's route config file directly imports a component or service from the other feature's folder instead of through a shared/common location; (c) a genuinely shared heavy dependency gets hoisted into one of the two chunks rather than a separate shared chunk, if the bundler's heuristics decide it's not worth a third split. Diagnosis: run `ng build --stats-json` and inspect with `source-map-explorer` or `webpack-bundle-analyzer` to see exactly which source files ended up in which chunk; grep import statements between the two feature folders for a direct cross-import; consider adding lint-enforced module boundaries (e.g., Nx module boundary rules or a custom ESLint import-restriction rule) to prevent this at the source rather than discovering it in bundle analysis.

Q12. What's the difference between lazy loading via the router and lazy loading via `@defer` blocks?
Router-based lazy loading (`loadChildren` / `loadComponent`) is navigation-driven — a chunk loads when the URL changes to match a route. `@defer` (Angular 17+) is a template-level, declarative mechanism for deferring the rendering (and thus the loading) of a specific piece of a **currently-rendered** view based on triggers like viewport visibility, user interaction, idle time, a timer, or a custom condition — independent of routing entirely. Both ultimately compile down to dynamic `import()` calls that the bundler splits into separate chunks, but `@defer` is for granular in-page lazy loading (e.g., a heavy comments widget below the fold on an otherwise-eager page), while router lazy loading is for whole route/page-level splitting.

Q13. Why might raw dynamic `import()` calls fail to be split into a separate chunk?
If the import specifier isn't a statically analyzable string literal — e.g., `import(getModulePath())` where the path is computed at runtime from a variable whose value isn't known at build time — the bundler cannot determine ahead of time which module(s) might be requested, so it either can't split it at all (falls back to bundling all statically-analyzable candidates matching a glob-like pattern, if using a limited template literal), or throws a build warning/error depending on bundler configuration. The specifier must be a literal string (or a very constrained template literal referencing a small enumerable set of files) for esbuild/Webpack to create a distinct chunk and generate the correct chunk-loading manifest entry.
Interviewer intent: verifies the candidate understands that "lazy loading" is fundamentally a static, build-time-analyzable contract, not a fully dynamic runtime capability — a subtlety that trips up engineers who try to build fully data-driven plugin-loading systems naively.

Q14. Can you have a circular dependency between two lazy-loaded feature areas, and what happens if you do?
You can write one (Feature A's routes file imports something from Feature B's folder, and Feature B imports something back from Feature A), but it defeats the purpose of splitting them: the bundler may be forced to merge the mutually-dependent parts into a combined chunk, or duplicate the shared portion into both chunks depending on its dependency-graph heuristics, either way bloating what should have been two small independent chunks. It can also cause partial-initialization bugs in production if a barrel `index.ts` file re-exports across both features and class field initialization order ends up differing between dev-server and production module evaluation order. The fix is architectural: shared code used by multiple lazy features must live in a common module both depend on eagerly (landing in a shared chunk or the main bundle), and feature areas should never import directly from one another — enforced via lint rules/module boundaries in CI.

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7. Quick Revision Cheat Sheet

- `loadComponent` → lazy single standalone component, no child routes at that node.
- `loadChildren` → lazy feature area; resolves to standalone `Routes` array (modern) or `NgModule` (legacy); supports nested children.
- Both are just `() => import('...').then(m => m.X)` — the router calls this only when the route is matched (or when preloaded).
- **Route** `providers: [...]` creates a new child `EnvironmentInjector` scoped to that route subtree — same duplicate-instance risk as legacy `NgModule.providers`. Use `providedIn: 'root'` for true singletons.
- **Preloading strategies:** `NoPreloading` (default, on-demand only) / `PreloadAllModules` (fetch everything shortly after first nav) / custom `PreloadingStrategy` (selective, via `route.data`, connection quality, etc.).
- Preloading fetches **JS only** — no guards, no resolvers, no injector creation, no component instantiation until real navigation.
- `RouterPreloader` kicks off right after the **first** `NavigationEnd`.
- Chunking is generic bundler behavior: any static, literal-string dynamic `import()` becomes a split point — router config or plain application code, doesn't matter.
- Non-routed lazy loading = plain `await import('lib')` inside a method/service; memoize the promise to avoid re-fetching.
- `@defer` (Angular 17+) = declarative, template/view-level lazy loading, complementary to but distinct from router-level lazy loading.
- Bundle analysis: `ng build --stats-json` + `source-map-explorer` / `webpack-bundle-analyzer`; watch initial bundle size and check lazy chunks aren't accidentally merged or duplicated.
- Circular lazy imports between feature areas → bundler can't split cleanly → move shared code to a common eagerly-loaded module; enforce with lint/module-boundary rules.
- `canMatch` can prevent a lazy chunk from ever being fetched for unauthorized/flagged-off users (runs before match); `canActivate` runs after match, so the chunk fetch is typically already underway.
- Dynamic import specifiers must be statically analyzable string literals — fully runtime-computed paths break code splitting.

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